

DATABASE SYSTEMS I

SQL server express edition

Tuesday 9th September 2023

- A database is a shared, integrated computer structure that stores a collection of end user data and metadata.
- End user data - raw facts of interest to end user.
- Metadata is ^{e.g. data type} data about data. ^{DBMS is intermediary between user and the database.}
- DBMS is a collection of programs that manages the database structure and controls access to data stored in the database.

Advantages of DBMS.

- Improved data security through data privacy and security policies.
- Better data integration due to wider access hence easier to see how actions in one segment of the company affect other segments.
- Minimized data ^{different versions of same record} inconsistency due to properly designed database.
- Improved data access. e.g. queries.
- Improved decision making due to well designed database.
- Improved data manipulation through formulas.

Types of databases ^{operational/transactional/production database is used to support day-to-day operations.}

- a Can be classified according to # of users, database location(s), and the expected type and extent of use. ^{Data warehouse stores data used to generate info required to make decisions.}
- Users - single and multi-user database.

Workgroup database - Supports relatively small number of users.
Enterprise database - Supports many users.

b Location - centralized and distributed database.

c Data structure ^{Extensible Markup Language (XML) is a special language used to represent and manipulate data elements in a textual format.}

Unstructured - exist in their original (raw) state

Structured - Formatting unstructured data.

Semistructured - Already processed to some extent.

- Database design refers to the activities that focus on the design of the database structure that will be used to store and manage end-user data.

Evolution of file system data processing.

- Data processing (dp) specialist was hired to create a computer-based system that would track data and produce required reports.
- Field - A character or group of characters that has specific meaning.
- Record - A logically connected set of one or more fields that describes an entity.
- File - Collection of related records.

Problems with file system data processing

- A critique of the file system method serves two major purposes:
 - a) Understanding the shortcomings of the file system enables one to understand the development of modern databases.
 - b) Many of the problems are not unique to file systems hence understanding the problem prevents duplication of the same in a DB.

1. Lengthy development times - Simple data-retrieval tasks required extensive programming.
2. Difficulty in getting quick answers - The need to write programs to produce even the simplest reports makes ad hoc queries impossible.
3. Complex system administration due to increased number of files as the system expands.
4. Lack of security and limited data sharing
5. Extensive programming is required when making changes.

Structural and data dependence

- A file system exhibits structural dependence, which means that access to a file is dependent on its structure.
- Structural independence exists when it is possible to

Data integrity is condition in which all of the data in the DB are consistent with real-world events and conditions.

make changes in a file structure without affecting the application program's ability to access the data.

- Data dependence is when data access programs are subject to change when any of the file's data storage characteristics change.

Data redundancy

- Exists when the same data are stored unnecessarily at different places.
- Uncontrolled data redundancy leads to;
 - a) Poor data security - Data becomes susceptible to unauthorized access.
 - b) Data inconsistency - Different and conflicting versions of the same data appear in different places. e.g. Peter K. & P. Karanja. Inconsistent data lacks data integrity, ^{accuracy} ^{verifiability}.
 - c) Data anomalies develop when not all of the required changes in the redundant data are made successfully. The types of anomalies;
 - Update anomalies -
 - Insertion anomalies
 - Deletion anomalies

Wednesday 10th September 2025

DATABASE SYSTEMS

- Refers to an organization of components that define and regulate the collection, storage, management and use of data within a database environment.

DBMS functions

1. Data dictionary management - Data dictionary stores definitions of the data elements and their relationships (metadata).
2. Data storage management - DBMS creates and manages the complex structures required for data storage.
3. Data transformation and presentation - Transforms data to conform to required data structures.
4. Security management - Creates a security system that enforces user

security and data privacy.

5 Database communication interfaces. Accepts end-user requests via multiple, different network environments.

Disadvantages of database systems:

1. Increased costs. Sophisticated hardware and software.
2. Management complexity. Interfere with many different technologies and have a significant impact on a company resources and culture.
3. Maintaining currency. One must keep their system current to promote efficiency.
4. Vendor dependence. Heavy reliance on a ^{vendor} company's resources makes it difficult to change vendors.
5. Frequent upgrades cost a lot of money.

DATA MODELS

- This is a simple graphical representation of more complex real world data structures.

Data model basic building blocks

- The basic building blocks are entities, attributes, relationships, and constraints.
- An entity is anything about which data are to be collected or stored e.g person, customer
- An attribute is a characteristic of an entity. e.g customer name
- A relationship describes an association among entities.
 - a) One-to-many (1:M or 1...*)
 - An entity instance of A can be associated with ~~at~~ most any number of entity instances of B; An entity instance of B can be associated with at most one entity instance of A.
 - b) One-to-One (1:1)
 - An entity instance of A can be associated with at most one entity instance of B; An entity instance of B can be

associated with at most one entity instance of A.

c) Many-to-many (m:n)

- An entity instance of A can be associated with any number of entity instances of B; an entity instance of B can be associated with any number of entity instances of A.

- A constraint is a restriction placed on data e.g. referential integrity.

Business rule

- Is a brief, precise description of a policy, procedure or principle within a specific organisation.

- An example is that a ^{noun} customer may ^{verb} generate many ^{noun} invoices.

- A noun in a business rule will translate to the entity in the model and the verb associating nouns will translate to the relationship.

- Business rules are essential in database design;

a) Standardize company's view of data

b) Communication tool between users and designers.

c) Allow designer to understand the nature, role and scope of data.

d) Allow designer to understand business processes.

e) Allow designer to create an accurate model by developing appropriate relationship participation rules and constraints.

Naming conventions

1. Entity names should be descriptive of objects in the business environment

2. Entity names use terminology familiar to the users.

3. Attribute name should be descriptive of the data it represents e.g. ^{cus_credit} - limit

Evolution of data models.

*1. Hierarchical model - contains levels or segments.

2. Network model - Represent complex data relationships more effectively

- Some important concepts were;

a) Schema - Conceptual organization of entire database as viewed by

And use People who use application programs to manipulate data and generate info.
- Logical design refers to creation of conceptual data model that could be implemented in any DBMS.

db administrator.

b) Subschema - Defines portion of db seen by application programs that actually produce desired info from data contained within db.

c) Data management language (DML) - Environment in which data can be managed and to work with data in the db.

d) A schema data definition language (DDL) - enables db administrator to define the schema components.

external model

↓

conceptual model

↓ logical independence

internal model

DBMS View

↓ physical independence

physical model

*external model.

Tuesday 16th September 2023

Degrees of data abstraction.

- The external model is the end user's view of the data environment.
- The conceptual model combines external models into one model that covers all user requirements, i.e. designer's view.
- A specific representation of an external view is known as external schema.

The conceptual model

- Represents a global view of the entire database by the entire organisation. Also called conceptual schema.
- The most commonly used conceptual model is ER model.
- The advantages of conceptual model include;
 - a) Provides a bird's eye view of the data environment that is relatively easy to understand.
 - b) It is both software and hardware independent.

The internal model

- This is a representation of the DB as "seen" by DBMS.
- An internal schema depicts a specific representation of an internal model using the database constructs supported by the chosen DB.
- Is software dependent and hardware independent.

logical independence is when one can change the internal model without affecting the conceptual model.

The physical model

- Operates at the lowest level of abstraction
- Software and hardware dependent.
- Requires the definition of both the physical storage devices and the access methods required to reach the data within those storage devices.

Tuesday 25th November 2025.

- Referential integrity restricts the user from creating orphan records. Orphan records can occur when a parent record containing child's records are deleted, or if parent records are updated hence mismatch in records ^{between parent & child.} or if one adds a child record that is linked to a non-existent parent.
- Cascade update option when creating relationships ensures that changes made in parent table reflect in child table.
- Cascade delete option ensures that incase records are deleted in parent table this deletion also reflects in child table.

Types of relationships

1. Unary - A table has a relationship with itself. e.g. one employee managing other employees.
2. Binary - Between two tables e.g. customer and orders
3. Ternary - A relationship among three tables e.g. student, exam, unit

Relationship cardinality.

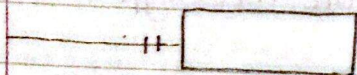
- Degree of a relationship.



Optional many



Mandatory many.



Mandatory one.

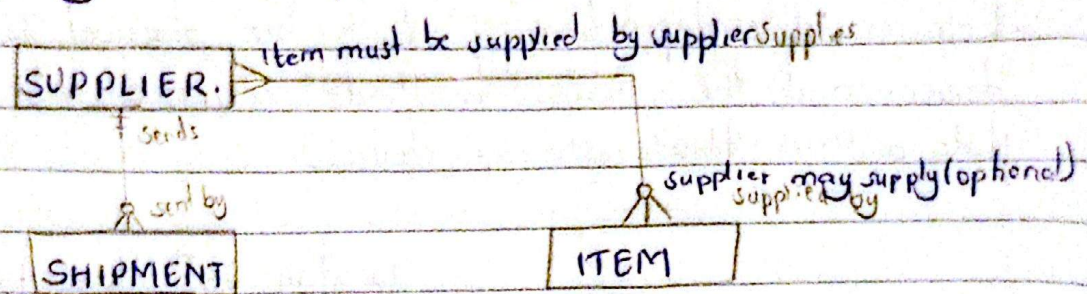


Optional one.

entity is a person, place, object or concept in the user environment about which the organization wishes to maintain data.

For example

A ^{co ps} supplier may supply many ITEMS. Each ITEM is supplied by any number of SUPPLIERS.



A ^{co ps} SUPPLIER may send many shipments. Each SHIPMENT must be sent by exactly one SUPPLIER.

Wednesday 26th November 2025

Modelling entities and attributes

- A strong entity type is a ~~collection~~ of one that exists independent of other entity type. e.g student.

- Entity type/entity instance

Entity type is a collection of entities that share common properties while entity instance is a single occurrence of an entity type.

- A weak entity type is an entity type whose existence depends on some other entity type. e.g DEPENDENT in employee table.

- Required attributes - An attribute that must be present for each entity instance whereas an instance that may not have a value is called optional attribute.

- A composite attribute is an attribute such as address that has meaningful component parts which are more detailed attributes.

- A simple/atomic attribute is one that cannot be broken down into smaller components that are meaningful to the organization.

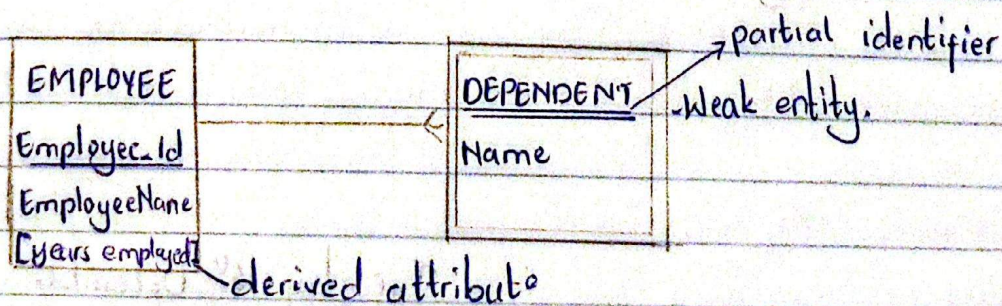
- A single-valued attribute contains one value while multivalued

attributes may take more than one value.

- A derived attribute is one whose values can be calculated from related attribute values. Mostly indicated in an ER diagram using square brackets around attribute name.

Modelling relationships.

- A relationship is an association representing an interaction among the instances of one or more entity types that is of interest to the organization.
- Relationship instance is an association between (or among) entity ~~types~~ instances, where each relationship instance associates exactly one entity instance from each participating entity type.
- Associative entity is an entity type that associates the instances of one or more entity types and contains attributes that are peculiar to the relationship between those entity instance.



Multivalued attributes

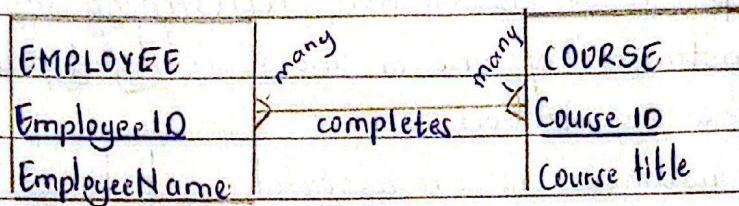
Must specify the separator e.g. (;). For example for skill e.g. C++, Java and HTML we can write them as C++;Java;HTML;SQL.

- It isn't recommended to write derived attribute in the table for example age. If it must be added into the ER diagram put it in squared brackets.

Residual attribute
 Is attribute that doesn't fit in either ^{side} of the main relationships but fits in the associative entity.

Modelling relationships

- A relationship is an association representing an interaction among the instances of one or more entity types that is of interest to the organization.
- A relationship type is a meaningful association between entity types and is denoted by a line labelled with the name of the relationship.



Associative entity.



Certificate is an associative entity hence represented using rounded rectangle.

external model

OODM

Network model

Data manipulation language

extended relational model

Tuesday 16th December 2025

Relational database model

Predicate logic provides framework for establishing assertions.

- Based on predicate logic and set theory.
- Based on these ~~some~~ components;
 - a) logical data structure represented by relations ie. data represented in rows and columns
 - b) Set of integrity rules to enforce that the data remain consistent over time. Assertion is a statement which will be either a true or a false
 - c) Set of operations that defines how data are manipulated.
- Set theory forms basis for data manipulation.

A logical view of data

- The DBMS manages and controls access to the data and database structure, hence placing DBMS between application and database eliminating most of the file system's limitations.
- The result of such flexibility is a far more complex physical structure.
- The relational data model changed all of that by allowing the designer to focus on the logical representation of data and its relationships rather than physical storage details. e.g. automatic car instead of manual car.

Tables and their characteristics

- Relation is the logical construct used in creation of data relationships.
- A table contains a group of related entity occurrences i.e. entity set.
- A table that doesn't have characteristics of a relational table is called cursor.
- Entity set and table are often used interchangeably.

Characteristics of a relational table.

- a Two-dimensional structure composed of rows and columns.
- b Each row (tuple) represents a single entity occurrence within the entity set.
- c Every column represents an attribute and has a unique name.
- d Intersection of a column and a row represents a single value.
- e All values in a column must conform to the same data format.
- f Each column has a specific range of values known as attribute domain.
- g The order of the rows and columns is immaterial to the DBMS i.e. doesn't matter.
- h Each table must have an attribute or combination of attributes that uniquely identifies each row.
- i Every relational table must have a primary key.

-The data types supported in a DBMS are;
numeric, character, date and logical!

NB Sometimes rows are referred to as records, columns as fields and table as a file in some DBMS.

Keys - used to establish relationships among tables and to ensure integrity of data.

- Used to ensure that each row is uniquely identified.
- A key consists of one or more attributes that determine other attributes.
- Either key or non-key attributes.
- Key attributes determine non-key attributes. ie determinant $\rightarrow$ dependent.

Dependencies

- Keys use principle of determination:

a) Functional dependence

- If one uses the key attribute to determine the non-key attribute they only get one ^{occurrence} value ie $STU_NUM \rightarrow STU_LNAME$
 STU_NUM functionally determines STU_LNAME , and STU_LNAME functionally depends on STU_NUM .
- Full functional dependence refers to functional dependencies in which the entire collection of attributes in the determinant is necessary for the relationship.

Types of keys

a) Composite key - functional dependence.

A key that has more than one attribute eg
 $stu_fname, stu_lname, stu_dob \rightarrow stu_email, stu_phone$
 $stu_no, stu_dob \rightarrow stu_fname, stu_lname$

b Super key. functional dependence.

- Uniquely identifies each row in the table.

The composite keys above can also be super keys as the attributes ^{determined} uniquely exist in the table.

stu-no $\rightarrow$ stu-fname, stu-lname, stu-dob.

c Candidate key

- A minimal super key ie a super key without unnecessary attributes.

- Exhibits full functional dependence.

stu-no $\rightarrow$ stu-fname, stu-lname, stu-dob

stu-fname, stu-lname, stu-dob $\rightarrow$ stu-email, stu-phone

The most straightforward primary key from the candidate keys above is stu-no.

d Primary key.

- Is a unique identifier eg stu-no

e Foreign key

- Is a primary key in one table appearing as a normal field/attribute in another table to facilitate relationships.

f Secondary key.

- Mostly used for data retrieval purposes. eg using phone # to retrieve patient details.

Entity integrity

- Condition where each table has a primary key.

- Primary key requirements;

- a) Must be unique (all the values in the primary key)

- b) no key attribute in primary key can contain a null

- A null is the absence of value and is problematic

to the relational database model.

- We can use isnull function or the coalesce function that replaces the null with a value

- A null can represent;

 - a) Unknown attribute value e.g dob for older people

 - b) A known, but missing attribute value.

 - c) A "not-applicable" condition

- To manage nulls, we use flags to indicate the absence of some value.

- Flags are special codes that indicate absence of some value.

Integrity rules

1. Entity integrity.

- All primary key entries are unique and must not contain a null hence each row will have a unique identity and foreign key can properly reference primary key values.

2. Referential integrity

- Every non-null foreign key value must reference an existing primary key value.

3 NOT NULL

- Can be placed on a column to ensure that every row in the table has a value for that column.

4 UNIQUE

- Restriction placed on a column to ensure that no duplicate values exist for that column.